

Estimating Gaze From Head and Hand Pose and Scene Images for Open-Ended Exploration in VR Environments

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1 INTRODUCTION

The widespread utility of eye tracking technology has created a growing demand for more consistent and reliable eye-tracking systems, and there is a need for new and accessible approaches that can enhance the accuracy of eye-tracking data. Previous studies have offered evidence for associations between certain non-eye signals and gaze such as a strong coordination between head motion and gaze shifts, e.g. [3], hand and eye spatiotemporal statistics, e.g. [7], and gaze behavior and scene content, e.g. [2]. Previous studies have also shown how various combinations of eye, head, scene, and hand signals can be leveraged for applications such as gaze estimation [5, 10], prediction [8], and classification [6]. Though these previous approaches provide support for the idea that non-eye sensors (i.e. head, hand, and scene) are useful for estimating gaze, they have not yet fully addressed how these signals individually and in combination contribute to gaze estimation.

To address this, we collect a large-scale dataset of sequences of head, hand, scene, and gaze signals sampled at 90Hz as users freely explore indoor and outdoor virtual environments. We use a machine learning framework (i.e., gradient boosting regression trees (GBRT) [1]) trained by this dataset to quantify the contribution to gaze estimation of each individual non-eye sensor (head, hand, and scene) and the combination thereof. We show that all the sensor models exhibit improved performance over baseline models across all cross-validation methods, and thus conclude that head, hand, and scene signals comprise useful information that can complement traditional eye-image, sensor-based eye trackers to inform gaze estimation models.

2 RECORDING HEAD, HAND, SCENE AND GAZE DATA

Participants: The dataset was captured from 44 voluntary participants (age (years) = 31.7 (SD 10.5); 20 females; 40 right-handed; ipd (mm) = 62.95 (SD 3.62)).

Apparatus: Participants wore an in-house prototype virtual reality head-mounted display (VR-HMD) and used Oculus Rift sensors and controllers for interaction. We recorded the head pose using Oculus Rift's inertial measurement unit (IMU) at a sampling rate of 90 Hz and used the Unity game engine to display the scenes.

Stimuli: We created one indoor and outdoor scene each with the same interactive content that aimed to capture a wide range of gaze behaviors (Figure 1). The interactive content included a table of graspable objects (including a clipboard of instructions and a gun which initiates a shooting game) and a canvas with crayons and an eraser. Together these objects provided the opportunity for many



Figure 1: Examples of the indoor and outdoor scenes and activities (a) holding an object; b) playing a shooting game; c) drawing; d) reading) during VR gameplay. The 3D gaze vector has been shown in these images for visualization purposes (green dot), though it was not presented during the experiment.

behaviors such as reading, throwing, object-manipulation, drawing, aiming, and shooting.

Procedure: Participants were asked to explore the environments (indoor and outdoor) for up to five minutes each with no specific task, and they could quit participation when they wished.

3 GAZE ESTIMATION

We used a machine learning framework to explore whether the recorded non-eye sensors (head, hand, and scene), individually and together, are informative for estimating gaze.

3.1 Feature engineering

All recorded signals were referenced to the center pupil frame (i.e. center of the head between the eyes) making the dataset egocentric. The axis orientation is represented in Figure 1a. Each sensor signal was measured at 90Hz for a total of 2,194,865 samples of data. To prepare our dataset for training gaze estimation models, we discarded all frames with missing gaze vectors and outlying ground truth gaze vectors (i.e. with greater than 99 percentile scores), reducing the dataset to 1,933,551 frames. We then created a data subset that required an equal number of frames from each user and each scene type resulting in a final dataset of 31 users each with 2.5 minutes of game play from each scene for a total of 837,000 frames.

Head: We included head orientation as input to our gaze estimation model. The head orientation was represented by a unit quaternion at each frame (specifically X , Y , Z , and W coordinates)

Hand: The left and right hand orientations and positions were included as input to our gaze estimation model. The left and right hand orientations were represented as a unit quaternion at each frame (X , Y , Z , and W coordinates). The position of each hand relative

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k -fold method	Baseline Models			Sensor Models			
	random	center	mean	head	hand	scene	all
Random	30.11 (0.025)	10.07 (0.004)	9.36 (0.006)	5.32 (0.007)	2.49 (0.013)	3.70 (0.012)	2.89 (0.014)
Subj-stratified	30.12 (0.022)	10.07 (0.015)	9.36 (0.008)	5.30 (0.019)	2.46 (0.016)	3.58 (0.010)	2.86 (0.007)
Subj-independent	30.12 (0.210)	10.04 (0.575)	9.35 (0.527)	8.51 (0.389)	8.81 (0.327)	8.21 (0.279)	8.05 (0.257)

Table 1: Baseline and sensor model performances for gaze estimation reported in terms of the average and standard deviation (in parentheses) of the angular error (in degrees) between predicted and ground truth 2D gaze angles for each cross-validation method.

to the center of the head in meters (X , Y , and Z coordinates) was included.

Scene: Scene images were originally recorded as the on-screen image presented to the user at each frame. These were stored as 3D color images (RGB) at a pixel resolution of 128 x 71. We created visual saliency maps (i.e. expected density map of eye fixations given the content of the scene) for each image using a pre-trained, state-of-the-art SAM-ResNet saliency predictor [2]. Due to computational constraints, we down-sampled each saliency map to 32x18 pixels for training.

Annotations: Ground truth 3D gaze vectors in meters are provided for each frame. We obtained gaze vectors and cornea centers of each participant using a user-calibrated glint-based model [4]. We transformed the original ground truth 3D gaze vectors to 2D gaze angles, (θ, ϕ) , for training and testing.

3.2 Method

Sensor Models: To address whether the non-eye sensors are more informative than our baseline models (see below) for the task of gaze estimation, we trained a separate gradient boosting regression tree (GBRT) model [1] for each non-eye sensor i.e., head, hand, scene, and one early fusion model combining all these sensors with an equal weighting of importance [9]. For each GBRT model, we adopted a grid-search hyper-parameter optimization strategy to estimate the relevant hyper-parameters: learning rate, number of trees, tree depth, child weight, minimum loss reduction required for partitioning γ , and the L1 regularization term on the weights α .

Baseline Models To evaluate whether the non-eye sensors were informative for gaze estimation, we compared the performance of the sensor models to each of three baseline models, i.e. random, subject-stratified, and subject-independent. The *random baseline model* estimates the 2D gaze angles directly from a random normal distribution truncated by the field-of-view of the HMD. Our *center baseline model* estimates the 2D gaze angle as the center point (i.e. $Y(0,0)$) for each frame. Our *mean baseline model* estimates the 2D gaze angle as the mean gaze vector of the training set.

Cross-validation: We instituted three different methods for the five-fold cross-validation procedure. For the first, the *Random* method, we created the five groups by randomly shuffling the original data and creating five equally-sized subsets. For the second, the *Subject-stratified* method, we required that each group contain an equal representation of data from each subject. For the third, the *Subject-independent* method, we ensured that each of the five groups included data from a unique set of subjects. We cross-validated all baseline and sensor models by all three methods.

Evaluation Metric: We reported the gaze estimation results in terms of the spatial accuracy of the gaze estimate for each model. We defined spatial accuracy as the mean angular error in degrees between the estimated and ground truth 2D gaze angles.

3.3 Results

Table 1 and Figure 6 show that the gaze estimation model error for each individual non-eye sensor (head, hand, and scene) and an early fusion model combining all non-eye sensors outperform each of the baseline models across all three cross-validation methods. From

these analyses, we conclude that the measured non-eye sensors are informative for gaze estimation. However, the experiments we have reported are only baseline methodologies that support the further utility of this dataset for creating more complex and robust gaze estimation models with non-eye sensor data and other applications that could be informed by the relationships between these signals.

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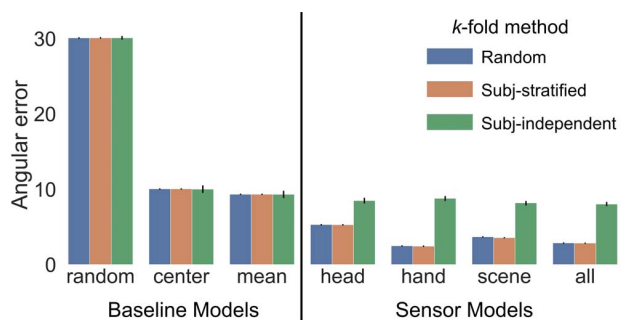


Figure 2: The spatial accuracy (i.e. angular error in degrees) in the gaze estimate for baseline and sensor GBRT models (error bars indicate 1 SEM).